Anti-Corruption Base Intelligence

Software Requirements Specification (SRS)

Plantel: Centro Universitario de los Valles

Materia: Software Configuration Management

Nombre del profesor: *Omar Ali Zatarain Durán*

Grado: Tercer semestre

Prepared by: Hiram Gabriel Alonso Marquez

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Software Requirements Specification (SRS) for AcBi

1. Introduction

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements for "Paradox Mobile," a cross-platform mobile application designed to assist investigators, cheack the Table 1. Definitions for handling definitions.

1.1 Definitions, Acronyms, and Abbreviations

Term	Definition
API	Application Programming Interface – Interfaces for data exchange with external public sources.
E2EE	End-to-End Encryption – Encryption ensuring data is secure from source to destination.
RTL	Relational Time Lines – Chronological views comparing official and captured data.
SHA-512	Secure Hash Algorithm 512 – A cryptographic hash function for data integrity.
Tor	The Onion Router – A network for anonymous communication.
VPN	Virtual Private Network – A secure tunnel for internet traffic.
Noise Threshold	A configurable metric for detecting inconsistencies in data timelines.

Term	Definition
Honeypot Inverso	A defensive mechanism generating fake data upon intrusion detection.
Local-First	Design where all processing occurs on the device, with optional external access.

Table 1. Definitions

2 Product Perspective

It is a novel tool in the domain of anti-corruption technology.

2.1 Product Functions

The application enables users to perform the following:

- Capture and monitor public data changes (e.g., new companies, financial movements, geolocations) via configurable APIs.
- Build and visualize dynamic graphs of relationships between officials, families, and entities, with risk scoring.
- Track full lifecycles of bureaucratic processes, flagging anomalies like ignored bids or accelerated timelines.
- Generate relational timelines comparing official declarations to captured signals, calculating corruption probabilities.
- Provide a central dashboard for oversight.

2.2 User Classes and Characteristics

- Primary Users: Investigators and detectives – Experienced analysts requiring forensic tools; assume technical proficiency in data interpretation but not in programming.

3. Specific Requirements

3.1 Functional Requirements

50 Functionalities

Module 1: Data Listener

- API Integration
- Verification of Manipulated APIs
- Metadata Capture
- Listening for changes in the Public Registry of Commerce (new companies linked to officials).
- Detection of bank account openings or public financial transactions of family members.
- Monitoring of changes in properties and vehicles registered in the names of family members or acquaintances.
- API Editing: DELETE, EDIT
- Manipulated API Report
- OSINT INTEGRATION
- API MANAGER.

Module 2: Network Analysis

- Construction of dynamic relationship graphs (officials ↔ family members ↔ companies).
- Graph filtering by politician and time period.
- Massive data cross-referencing.
- Cross-referencing of projects with information from Module 1.
- Risk scoring by degree of closeness.
- Detection of "shared accounts" or transfers between close circles.
- Mapping of closed groups or chats where officials and beneficiaries appear in lists.
- Interactive visualization of graphs with zoom.
- Graph update button.
- Filters

Module 3: Process Tracking

- Tracing the lifecycle of the data collected for the case by day, in PDF table format.
- Tracing by name.
- Monitoring of unusual accelerations in bureaucratic deadlines.
- Historical record of each data collection by package.
- AUTOMATIC COMPARISON BETWEEN SUBMITTED OFFERS AND THE WINNING ONE, BY FILE NAME
- Tracing of beneficiaries identified in the network graph.
- Tracing of budget transfers between departments

- Tracing of approvals.
- Preparation of information in timelines that encompass Module 3
- Visual timeline by process with highlighted anomalies.

Module 4: Relational Timelines (RTL)

- Generation of relational timelines by politician/institution/project.
- Calculation of the "noise threshold" (inconsistencies between declared and observed information).
- Temporal comparison: official statements vs. captured signals.
- Detection of approvals before official requests (Basic Time Paradox).
- Algorithms to predict the probability of corruption based on historical noise (Time Paradox).
- User-configurable threshold (e.g., >25% noise = high alert).
- Comparative visualization of timelines (official vs. observed).
- Detection of unexplained "skips" in the authorization flow.
- Recording of specific contradictions (e.g., "politician claims to be unaware of the company, but a family member is a partner").
- Export of RTLs as PDFs

Module 5: Time Paradox

- Main dashboard with filters by date, politician, institution, and project.
- Chronological display of all RTLs with noise thresholds.
- Push/email/SMS alerts when the noise threshold is exceeded.
- Automatic flag: "Start Investigation Case" with collected evidence.
- Interactive risk map by state/municipality.
- Bulk export of reports for NGOs, journalists, or authorities.
- "Paradox" mode: highlights serious temporal inconsistencies in red.
- History of detected cases.
- Interactive interface with animations.
- Timeline with information sheets.

3.2 Non-Functional Requirements

- Data Security: All collected and processed data must be encrypted at rest and in transit.
- Forensic-Level Security: End-to-end encryption (E2EE).
- Immutable Digital Chain of Custody: Every piece of data captured, analyzed, or exported must have a certified SHA-512 hash and timestamp.

- Operational Anonymity of the Investigator: Native support for VPN/Tor + automatic IP rotation.
- Counterintelligence Resistance: Automatic detection and blocking of tracking attempts.
- Offline Availability.
- Scalability for Large Cases.
- Tolerance for Outages.
- Professional and Technical Interface: Dashboard designed for analysts.
- Protection Against Evidence Tampering: Automatic detection of attempts to edit or inject false data.
- Sandbox Detection and Reverse Analysis: The system must self-destruct or enter 'false positive' mode if it detects that it is being analyzed in a virtual or debug environment.
- Code obfuscation: Binary code and scripts obfuscated with multiple layers (VMProtect, Themida, or equivalent) to hinder reverse engineering.
- Memory dumping protection: Anti-memory dump mechanisms (anti-Process Explorer, anti-Cheat Engine, anti-DumpIt).
- Automatic key and certificate rotation: Daily generation and renewal of encryption keys and certificates.
- Keylogger and screen recorder detection: Automatic blocking or masking of sensitive input if capture tools are detected.
- Traffic analysis protection: Use of padding, timing randomization, and dummy traffic to hide communication patterns.
- Programmable self-destruction.
- Reverse honeypot mode: The system can generate false data or deceptive signals if it detects intrusion attempts.
- Internal researcher audit.
- API monitoring and encryption.

3.3 Functional Requirements Definition

RF-001 Integration with APIs (Type 1)

1. GENERAL INFORMATION

Field	Description
Requirement ID	RF-001
Name	Integration with Public Governmental APIs (Type 1)
Module	Module 1: Data Listener
Priority	High (Critical)
Status	Proposed
Version	1.0
Date	02/02/2026

2. DESCRIPTION

2.1 General Description

The system must allow the integration, configuration and consumption of public Mexican governmental APIs to capture data related to bureaucratic processes, tenders, budgets and asset declarations. Type 1 APIs are those officially provided by government institutions for public information consultation.

Process:

```
User configures API
   ↓
Click "Test Connection"
   ↓
System executes request via Tor/VPN
   ↓
Successful response?
   ├─ YES → Parse data → Show preview → Enable "Save"
   └─ NO  → Show error → Show suggestions → Disable "Save"
```

Inputs:

- Descriptive name of the API
- Base URL of the endpoint
- Query parameters
- Authentication type (None, API Key, Bearer Token)

- Credentials (if applicable, encrypted)
- Update frequency (manual, hourly, daily, weekly)

Outputs:
- Format
- Mandatory fields
- No duplication of already configured endpoints
- All configurations are stored encrypted locally (AES-256-GCM)

Constraints:
API credentials

Source: Government

RF-002: Detection of Manipulated / Compromised APIs

1. GENERAL INFORMATION

Field	Description
Requirement ID	RF-002
Name	Detection of Manipulated APIs
Module	Module 1: Data Listener
Priority	High (Critical)
Status	Proposed
Version	1.0
Date	02/02/2026

2. DESCRIPTION

2.1 General Description

The system must automatically detect whether a governmental API has been compromised, manipulated or is returning suspicious data through cryptographic integrity validations, heuristic response analysis, and comparison against historical patterns. This requirement is critical to guarantee the reliability of collected evidence and to protect researchers from falsified or maliciously injected data.

2.2 Process

```
System performs API capture
  ↓
SSL Certificate Validation
  ├─ Is the certificate valid and known?
  │   ├─ YES → Continue
  │   └─ NO  → CRITICAL ALERT → Pause capture
  ↓
Response Schema Validation
  ├─ Does the structure match historical patterns?
  │   ├─ YES → Continue
  │   └─ NO  → WARNING ALERT → Log anomaly
  ↓
Heuristic Data Analysis
  ├─ Are values within expected ranges?
  │   ├─ YES → Continue
```

```
   │   └─ NO  → MEDIUM ALERT → Flag suspicious records
   ↓
Data Volume Verification
   ├─ Is the volume consistent with historical data?
   │   ├─ YES → Continue
   │   └─ NO  → LOW ALERT → Log for review
   ↓
Reliability Score Calculation
   ├─ Score > 80    → Data TRUSTWORTHY
   ├─ Score 50–80   → Data SUSPICIOUS (requires review)
   └─ Score < 50    → Data NOT TRUSTWORTHY (automatically reject)
   ↓
Storage with Reliability Metadata
```

2.3 Inputs

Input	Type	Description	Mandatory
HTTP Response	Object	Full API response including headers, body and status	Yes
SSL Certificate	X.509	Server digital certificate	Yes
Historical Schemas	JSON	Previously validated response structures of this API	Yes (after 1st capture)
Historical Data	Array	Statistical sample of previous captures for heuristic analysis	No (improves accuracy)
Threshold Configuration	Object	Custom detection thresholds (volume, values, etc.)	No (uses defaults)
List of Known Certificates	Array	SHA-256 of valid SSL certificates from government entities	Yes

2.4 Outputs

Output	Format	Description
Reliability Score	Integer (0-100)	Score of confidence in the integrity of captured data

2.5 Mandatory Fields

- Valid server SSL certificate
- SHA-256 hash of the certificate
- JSON/XML structure of the response

- SHA-512 hash of the response body
- Capture timestamp (ISO 8601)
- API configuration ID (FK)
- Result of each validation (PASS/FAIL)

2.6 Constraints / Restrictions

Constraint	Description
Accuracy	False positive rate < 5% in normal conditions

2.7 Sources

- Government: SSL certificates of government entities (obtained from first manually verified capture)
- User: Custom detection thresholds according to the investigation case
- System: Historical capture records for statistical analysis

RF-003: Extended Metadata Capture

1. GENERAL INFORMATION

Field	Description
Requirement ID	RF-003
Name	Extended Metadata Capture
Module	Module 1: Data Listener
Priority	High
Status	Proposed
Version	1.0
Date	02/02/2026

2. DESCRIPTION

2.1 General Description

The system must automatically capture extended metadata from every request to governmental APIs, including network context, temporal, technical, and audit information. These metadata are essential to establish a digital chain of custody, validate the authenticity of evidence, detect manipulations, and provide complete forensic context for corruption investigations.

2.2 Process

Start
↓
Pre-request metadata capture
• Time, session, environment
↓
HTTP Request
↓
Response metadata capture
• Headers, status, network, SSL
↓
Data processing
• Parsing + hashes
↓
Chain of custody
• Signing + linked hash
↓
Secure storage

↓
Indexing and auditing

2.3 Inputs

Input	Type	Description	Mandatory
Full HTTP Response	HTTPResponse	Complete HTTP response object including headers, body, cookies	Yes
API Configuration	Object	Active API configuration (URL, method, params, etc.)	Yes
Device Status	Object	Battery, network, memory, available storage info	Yes
Network Context	Object	Tor circuit, active VPN, exit node, latency	Yes
Executing User	String	Anonymized hash of the user/detective performing the capture	Yes
Investigation Case	String (UUID)	ID of the associated investigation case (if applicable)	No
Start Timestamp	DateTime	Request start timestamp (UTC + local)	Yes
End Timestamp	DateTime	Process completion timestamp (UTC + local)	Yes
App Version	String	Version of AcBi running (e.g. "0.0.001")	Yes

2.4 Outputs

Output	Format	Description
Complete Metadata	JSON	JSON object containing all captured metadata, structured by categories

2.5 Mandatory Fields

Time Metadata:
- timestamp_start_utc (ISO 8601)
- timestamp_end_utc (ISO 8601)
- timestamp_start_local (ISO 8601 with timezone)
- timestamp_certified (RFC 3161)

2.6 Restrictions

Restriction	Description
Immutability	Metadata CANNOT be modified after initial storage

2.7 Sources

- Operating System: Device information (battery, network, memory)
- HTTP Server: Headers, cookies, SSL certificates
- Tor/VPN: Exit node, circuit, latency
- User: Investigation case ID (optional)
- Application: Version, configuration, parsing method used
- Network: Timestamp servers for RFC 3161 certification

RF-004: Monitoring Changes in the Public Commercial Registry

1. GENERAL INFORMATION

Field	Description

Requirement ID RF-004
Name Monitoring Changes in the Public Commercial Registry (New Companies Linked to Public Officials)
Module Module 1: Data Listener
Priority High
Status Proposed
Version 1.0
Date 02/02/2026

2. DESCRIPTION

2.1 General Description

The system must automatically monitor Mexico's Public Commercial Registries (RPC) to detect the creation, modification, or dissolution of companies that may be linked to public officials under investigation. This functionality enables the identification of potential conflicts of interest, shell companies, hidden beneficiaries, and corruption networks through the analysis of corporate formation patterns and their relationship with officials and their close circles.

2.2 Process

Monitoring of RPC is initiated by configuring target states, officials, keywords, frequency, and retrospective period.
The system periodically queries state-level RPC APIs and portals to obtain new incorporation deeds.
Records are filtered by date and relevant information is extracted: partners, legal representatives, corporate purpose.
Captured data is normalized: name standardization, RFC and address cleaning, format validation.
Duplicate records are detected and removed to ensure dataset consistency.
Normalized information is compared against target officials using exact matching, fuzzy matching, and cross-referencing by RFC, CURP, and address.
Possible family relationships are also analyzed through surname matching.
If no match is found, the record is archived without further processing.
If a match exists, a risk analysis is performed considering dates, corporate purpose, capital, common partners, and tender context.
If the calculated risk is low, the event is logged only as reference.

If the risk exceeds the defined threshold, a high-risk alert is generated, the user is notified, and a case is created.
Finally, the case is enriched with additional information: tenders, asset declarations, network analysis, and cross-referenced timelines.

2.3 Inputs

Input	Type	Description	Mandatory
List of Target Officials	Array	List of officials to monitor (full name, RFC, CURP, position)	Yes
States to Monitor	Array	List of Mexican states (e.g. ["CDMX", "Jalisco", "Nuevo León"])	Yes
Retrospective Period	Integer	Days to look back (e.g. 90 days)	Yes
Query Frequency	Enum	daily, weekly, biweekly, monthly	Yes
Additional Keywords	Array	Surnames, known family member names	No
Matching Thresholds	Object	Similarity configuration (fuzzy threshold, etc.)	No (uses defaults)
RPC Endpoints by State	Object	URLs of state RPC APIs/portals	Yes
Access Credentials	Object	API keys or credentials for RPC (if required)	No

2.4 Outputs

Output	Format	Description
Detected Companies	Array	List of new/modified companies with match to officials
Risk Score	Integer (0-100)	Corruption risk score per detected company
Generated Alerts	Array	HIGH/CRITICAL alerts for high-risk companies
Identified Relationships	Graph	Graph of relationships: official ↔ company ↔ partners
Event Timeline	Array	Chronological events: incorporation → tender → award
Findings Report	PDF/JSON	Detailed report with evidence and analysis
Integration to Module 2	Object	Data for Network Analysis (nodes, edges)

2.5 Mandatory Fields (MINIMUM 3)
Captured Company Data:
- razon_social (legal company name)
- rfc_empresa (company RFC)
- fecha_constitucion (incorporation deed date)
- estado (federal entity of the RPC)
- folio_mercantil (commercial registry folio number)
- capital_social (declared share capital)
- objeto_social (description of corporate purpose)

Partners/Shareholders Data:
- nombre_completo (partner full name)
- rfc_socio (partner RFC)
- curp (CURP of partner, if available)
- porcentaje_participacion (percentage of shares)
- tipo_socio (natural/legal person)
- es_representante_legal (boolean)

Detection Metadata:
- fecha_deteccion (detection date)
- matching_method (exact, fuzzy, RFC, etc.)
- matching_score (0-100, match confidence)
- funcionario_matched (official matched)
- risk_score (risk score 0-100)

2.6 Restrictions

Restriction	Description
Availability of APIs and OSINT sources	Agreement

2.7 Sources
- Government: State Public Commercial Registries (SIGER, RPPCDMX, etc.)
- User: List of target officials and monitoring configuration
- Module 2: Existing relationship network to enrich context
- Module 3: Active tenders for temporal cross-referencing

RF-005: Detection of Bank Account Openings and Public Financial Movements of Relatives

1. GENERAL INFORMATION

Field	Description
Requirement ID	RF-005
Name	Detection of Bank Account Openings or Public Financial Movements of Relatives
Module	Module 1: Data Listener
Priority	High
Status	Proposed
Version	1.0
Date	02/02/2026

2. DESCRIPTION

2.1 General Description

The system must automatically monitor public financial data sources in Mexico to detect bank account openings, unusual financial movements, asset transfers, and changes in tax declarations involving public officials under investigation, their direct relatives (spouse, children, parents, siblings), and close circles (godparents, known partners). This functionality enables the identification of illicit enrichment, money laundering, patrimonial conflicts of interest, and concealment of resources through the analysis of anomalous financial patterns and asset changes incompatible with declared income.

2.2 Process

Start of Financial Monitoring

↓

CONFIGURATION OF MONITORING

- Target officials and relatives
- Public financial data sources
- Query frequency
- Patrimonial anomaly thresholds
- Historical analysis period

↓

CAPTURE OF PUBLIC FINANCIAL DATA

- Asset declarations
- Modification declarations
- Published sanctions/fines
- Public tax liens/debts

```
  • Government contracts and payments
  ↓
NORMALIZATION AND STRUCTURING
  • Standardize amounts (to MXN)
  • Normalize dates (ISO 8601)
  • Clean RFCs and CURPs
  • Classify asset types
  • Validate data consistency
  ↓
MATCHING WITH OFFICIALS AND RELATIVES
  • Identify declarant (official)
  • Cross-reference with known relatives
  • Matching by RFC/CURP
  • Detect spouses and dependents
  ↓
Match found?
  ├─ NO  → [Archive]
  └─ YES → ANALYSIS OF ANOMALIES
           • Patrimonial growth
           • Income incompatibility
           • Undeclared assets
           • Family transfers
           • Timing with tenders
           ↓
      Anomaly > Threshold?
        ├─ NO  → [Register] [Normal]
        └─ YES → GENERATE ALERT
                 • HIGH alert
                 • Notify user
                 • Create case
                 • Financial timeline
                 ↓
             ENRICHMENT
               • Cross-reference tenders
               • Analyze family network
               • Calculate variation
               • Detect patterns
```

2.3 Inputs

Input	Type	Description	Mandatory
List of Target Officials	Array	Officials to monitor (name, RFC, CURP, position, agency)	Yes
List of Relatives	Array	Known relatives (name, RFC, CURP, relationship to official)	No
Financial Data Sources	Object	URLs of asset declarations APIs, public SAT, CNBV, etc.	Yes
Query Frequency	Enum	monthly, quarterly, yearly (declarations are periodic)	Yes
Historical Analysis Period	Integer	Years back to analyze evolution (e.g. 3 years)	Yes
Anomaly Thresholds	Object	Configuration of suspicious growth %, income incompatibility, etc.	No (uses defaults)
Asset Types to Monitor	Array	["real estate", "vehicles", "accounts", "investments", "companies"]	Yes
Access Credentials	Object	API keys for platforms requiring authentication	No

2.4 Outputs

Output	Format	Description
Detected Patrimonial Changes	Array	List of changes in assets of officials/relatives
Financial Anomaly Score	Integer (0-100)	Suspicion score of illicit enrichment

2.5 Mandatory Fields

Captured Patrimonial Declaration Data (MINIMUM 3):

* declarante_nombre (full name of the official)
* declarante_rfc (official's RFC)
* declarante_cargo (current position)
* declarante_dependencia (institution where they work)
* tipo_declaracion (initial, modification, conclusion)
* fecha_declaracion (submission date)
* ejercicio_fiscal (declared fiscal year)
* ingreso_neto_anual (total declared net income)

* activos_totales (sum of all assets)
* pasivos_totales (sum of all liabilities)
* patrimonio_neto (assets - liabilities)

2.6 Restrictions

Restriction	Description
OSINT	NO access to private banking information protected by banking secrecy

2.7 Sources

* Federal Government: Declaranet (National Transparency Platform)
* Federal Government: SAT - Public taxpayers registry
* Federal Government: CNBV - Published sanctions
* Federal Government: CompraNet - Amounts paid to suppliers
* State Governments: State transparency portals
* User: List of target officials and known relatives
* Module 2: Relationship network to identify suspicious transfers
* Module 3: Awarded tenders for temporal cross-referencing

RF-006: Monitoring Changes in Properties and Vehicles Registered to Relatives or Known Persons

1. GENERAL INFORMATION

Field	Description
Requirement ID	RF-006

Name Monitoring Changes in Properties and Vehicles Registered to Relatives or Known Persons
Module Module 1: Data Listener
Priority Medium-High
Status Proposed
Version 1.0
Date 02/02/2026

2. DESCRIPTION

2.1 General Description

The system must automatically monitor public real estate records (Registro Público de la Propiedad - RPP) and vehicle registries (REPUVE) in Mexico to detect acquisitions, transfers, sales, or changes of ownership involving public officials under investigation, their direct relatives (spouse, children, parents, siblings), and close circles (known partners, godparents). This functionality enables the identification of suspicious asset transfers, use of straw men to conceal assets, acquisitions incompatible with declared income, and patterns of asset concealment through comparative analysis of public records, asset declarations, and temporal context with awarded tenders.

2.2 Process

Start of Properties and Vehicles Monitoring
↓
CONFIGURATION OF MONITORING
- Target officials and relatives
- States/municipalities to monitor
- Types of assets (real estate, vehicles)
- Query frequency
- Retrospective period (e.g. 2 years)
- Suspicious value thresholds

↓
CAPTURE OF PUBLIC REGISTRY DATA
- Query RPP (Property Registry)
- Query REPUVE (Vehicle Registry)
- Filter by target names/RFCs
- Extract deed/card data
- Capture movement dates

```
  • Identify type of operation
  ↓
NORMALIZATION AND STRUCTURING
  • Standardize addresses
  • Normalize amounts (to MXN)
  • Clean RFCs and names
  • Classify property types
  • Validate market values
  • Detect duplicates
  ↓
MATCHING WITH OFFICIALS AND RELATIVES
  • Identify owner/holder
  • Cross-reference with target officials
  • Cross-reference with known relatives
  • Matching by RFC/CURP/name
  • Identify co-owners
  ↓
Match found?
  ├─ NO  → [Archive]
  └─ YES → ANALYSIS OF ANOMALIES
            • Incompatible value
            • Timing with tender
            • Family transfer
            • Not declared (RTL)
            • Straw man pattern
            ↓
      Anomaly > Threshold?
        ├─ NO  → [Register] [Normal]
        └─ YES → GENERATE ALERT
                  • HIGH alert
                  • Notify user
                  • Create case
                  • Visual timeline
                  ↓
              ENRICHMENT
                • Cross-reference declarations
                • Analyze property network
```

- Crossed timeline
- Calculate appreciation

2.3 Inputs

Input	Type	Description	Mandatory
List of Target Officials	Array	Officials to monitor (name, RFC, CURP, position)	Yes
List of Relatives	Array	Known relatives (name, RFC, CURP, relationship)	No
States to Monitor	Array	Mexican states to query RPP (e.g. ["CDMX", "Jalisco"])	Yes
Types of Assets	Array	["real estate", "vehicles"]	Yes
Query Frequency	Enum	monthly, quarterly (registries update slowly)	Yes
Retrospective Period	Integer	Years back to search for changes (e.g. 2 years)	Yes
Value Thresholds	Object	Minimum value of assets considered suspicious (e.g. properties >$500k)	No (uses defaults)
RPP/REPUVE Endpoints	Object	URLs of registry APIs/portals by state	Yes
Access Credentials	Object	Credentials if registries require authentication	No

2.4 Outputs

Output	Format	Description
Detected Assets	Array	List of new or modified properties/vehicles linked to officials/relatives
Patrimonial Anomaly Score	Integer (0-100)	Suspicion score of irregular transfer

2.5 Mandatory Fields

Captured Real Estate Data (MINIMUM 3):

* tipo_inmueble (house, apartment, land, commercial premises, warehouse, etc.)
* direccion_completa (street, number, neighborhood, postal code, municipality, state)
* coordenadas_gps (latitude, longitude - if available)
* superficie_terreno_m2 (land area in square meters)
* superficie_construccion_m2 (construction area in square meters)

* valor_catastral (official cadastral value in MXN)
* valor_operacion (declared purchase/sale value in deed)
* folio_real (folio number in RPP)
* numero_escritura (public deed number)
* fecha_escritura (deed signing date)
* fecha_registro (registration date in RPP)
* notario_numero (notary number)
* tipo_operacion (sale, donation, inheritance, adjudication, exchange)

2.7 Restrictions

Restriction	Description
OSINT	Unique source

RF-007: API EDITION - DELETE, EDIT

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-API-002
Name	API Edition - DELETE and EDIT functions
Module	API Management
Priority	High

Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION

System for complete API management that allows authorized users to delete and edit existing API configurations in the system catalog. Includes security validations, version control, and change auditing.

3. PROCESS

1. Access to the module: User logs in with authorized credentials

2. API selection: User searches and selects the API to modify/delete

3. EDIT:

* System displays form with current data
* User modifies allowed fields
* System validates changes
* System registers audit trail
* System applies changes and generates new version

4. DELETE:

* User requests deletion
* System checks for active dependencies
* System requests confirmation
* System performs logical/physical deletion
* System logs the action in audit trail

4. OUTPUTS

* Successful API update confirmation
* Successful API deletion confirmation

5. SOURCE

Government

6. RESTRICTIONS
Investigation case permissions

RF-008: MANIPULATED API REPORT

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-API-003
Name	Manipulated API Report
Module	API Audit and Reporting
Priority	High
Status	In definition
Version	1.0

Date 05/02/2026

2. DESCRIPTION
Reporting system that displays the complete history of manipulations (creation, editing, deletion) performed on system APIs. Allows tracking of changes and identification of responsible users.

3. PROCESS

1. Access to reporting module: Authorized user enters the reporting system

2. Filter configuration: User defines search criteria:

* Date range
* Operation type (creation, edit, deletion)
* Specific API or all
* User who performed the action
* Operation status (success/failed)

3. Report generation: System queries audit log and processes information

4. Visualization: System presents data in table/dashboard format

5. Export (optional): User downloads the report

4. OUTPUTS

* On-screen visual report with manipulation table
* PDF export

Details of each manipulation:

* Exact timestamp
* Responsible user

5. SOURCE
Investigator's research.

RTL Module.

6. RESTRICTIONS
Reports cannot be modified, deleted, or edited.

RF-009: OSINT INTEGRATION

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-OSINT-001
Name	Open Source Intelligence (OSINT) Integration
Module	Intelligence and Threat Analysis
Priority	High
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION
Integration system with open source intelligence (OSINT) sources to collect, analyze and correlate information to support requirements 4, 5 and 6.

3. PROCESS

1. Configuration of OSINT sources
2. Automated collection
3. Filters
4. Data enrichment

4. OUTPUTS
Normalized OSINT data feed for SIEM (Security Information and Event Management) and SOAR (Security Orchestration, Automation, and Response).

5. SOURCE

* Public OSINT sources

6. RESTRICTIONS
Applicable laws according to the investigation case.

RF-010: API MANAGER

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-API-001
Name	API Manager - Dashboard with Embedded Code Terminal
Module	API Management and Administration
Priority	Critical
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION

Centralized API management platform that integrates an interactive dashboard with an embedded code terminal. Enables complete lifecycle management of APIs (create, view, edit, delete, monitor) and provides real-time code execution and endpoint testing capabilities through an integrated terminal. Facilitates both visual and programmatic API management from a single interface.

3. PROCESS

1. Dashboard access:

* User authenticates with credentials
* System loads main dashboard with overall API overview

2. API visualization:

* Dashboard displays complete catalog of registered APIs
* Shows real-time metrics (usage, performance, status)
* Displays consumption graphs and statistics

3. API Management (graphical interface):

* Create new API via visual form
* Edit existing API configurations
* Delete APIs with confirmation
* Configure endpoints, authentication and permissions
* Manage versions

4. Integrated code terminal:

* User opens embedded terminal in the dashboard
* Executes commands for programmatic API management
* Performs real-time endpoint testing
* Runs batch configuration scripts
* Views logs and responses in console

4. OUTPUTS

* Interactive dashboard
* Code terminal

5. SOURCE
RF-001

RF-011: DYNAMIC RELATIONSHIP GRAPH CONSTRUCTION

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-GRAFO-001
Name	Dynamic Relationship Graph Construction (Officials ↔ Relatives ↔ Companies)
Module	Relationship Analysis and Data Intelligence
Priority	Critical
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION

Dynamic graph visualization and analysis system that allows identifying, mapping and analyzing connections between public officials, their relatives and associated companies. Uses network analysis techniques to detect patterns.

3. PROCESS

* System collects data from multiple sources
* Creates nodes for each identified entity
* Data enrichment
* Automated analysis

4. OUTPUTS
Differentiated nodes ready for visualization

5. SOURCES
Module 1

6. RESTRICTIONS
Availability of information collected from Module 1

RF-012: GRAPH FILTER BY POLITICIAN AND TIME

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-GRAFO-002
Name	Graph Filter by Politician and Time
Module	Relationship Analysis and Data Intelligence
Priority	High
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION

Advanced graph filtering system for relationship graphs that allows the user to refine and focus the visualization.

3. PROCESS

1. Access to the filters panel:

* User accesses the graph visualization module
* System displays filter panel with available options

2. Filter configuration

4. OUTPUTS

* Filtered graph

5. SOURCE

* Graph database with information

6. RESTRICTIONS
All relationships must have a start timestamp

RF-013: MASSIVE DATA CROSS-REFERENCING

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-DATA-001
Name	Massive Data Cross-Referencing
Module	Big Data Processing and Analysis
Priority	Critical
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION

Massive data processing and correlation system from multiple sources.

3. INTERNAL PROCESS

1. Configuration of data sources based on APIs and OSINT
2. Data extraction
3. Transformation and normalization
4. Cross-referencing process
5. Analysis and correlation
6. Validation

4. OUTPUTS
Data for RF-011

5. SOURCE
Module 1

6. RESTRICTIONS
Quality of the data found to perform the cross-referencing

RF-014: CROSS-REFERENCING PROJECTS WITH MODULE 1 INFORMATION

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-INTEG-001
Name	Cross-Referencing Projects with Module 1 Information
Module	Data Integration and Correlation
Priority	High
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION
Integration and correlation system that allows cross-referencing project information (contracts, tenders, investments, public works) with Module 1 data.

3. PROCESS

1. Extraction of project data
2. Extraction of data from Module 1 and RF-013
3. Normalization and preparation
4. Cross-referencing and correlation process
5. Pattern detection

4. OUTPUTS
Information for public projects filter

5. SOURCE

* Module 1 and RF-013

6. RESTRICTIONS
Information must be available from the sources

RF-015: RISK SCORING BY DEGREE OF PROXIMITY

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-RISK-001
Name	Risk Scoring by Degree of Proximity
Module	Risk Assessment and Scoring
Priority	Critical
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION
Automated risk scoring system that evaluates the probability of conflict of interest, corruption or irregularity based on the degree of proximity or closeness between entities (officials, relatives, companies, projects). Uses basic multifactorial scoring algorithms that consider: distance in the relationship graph (degrees of separation), type of link,

temporal context, economic magnitude and other risk factors to generate a numerical score that prioritizes cases for investigation and audit.

3. PROCESS

1. Identification of relationships

Definition of scoring parameters:

* Administrator configures weighting matrix considering:

  * Degree of proximity (0, 1, 2, 3+ degrees of separation)
  * Type of link (direct, family, corporate, etc.)
  * Official's role (senior management, middle, operational)
  * Involved amount (project, contract, transaction)
  * Temporal context (during term, before, after)
  * Sensitive sector (health, infrastructure, defense, etc.)

4. OUTPUTS

* Matrix score

5. RESTRICTIONS
Basic formula for a basic score

RF-016: DETECTION OF SHARED ACCOUNTS OR TRANSFERS BETWEEN CLOSE CIRCLES

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-DETECT-001
Name	Detection of Shared Accounts or Transfers between Close Circles
Module	Financial Analysis and Anomaly Detection
Priority	Critical
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION

Detects suspicious patterns of shared bank account usage and monetary transfers between individuals who are part of close circles.

3. PROCESS

Filters the output of RF-014 to search for patterns.

4. OUTPUTS
Description of the detected typology: circles

5. SOURCE
RF-014

6. RESTRICTIONS
RF-014 must be operational and delivering information.

RF-017: MAPPING OF CLOSED GROUPS WHERE OFFICIALS AND BENEFICIARIES APPEAR IN LISTS

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-CLUSTER-001
Name	Mapping of Closed Groups where Officials and Beneficiaries Appear in Lists
Module	Cluster Analysis and Network Detection
Priority	Critical
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION
The system identifies patterns of people appearing in multiple records.

3. PROCESS
An algorithm is built that identifies a person's name, lists people orbiting around them, and creates various conditions to search for them in RF-013.

4. OUTPUTS

Found data

5. SOURCE
RF-013 which provides the information

6. RESTRICTIONS

* Algorithmic – due to time limitations, the goal is to implement existing algorithms.

RF-018: INTERACTIVE GRAPH VISUALIZATION WITH ZOOM

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-VIS-001
Name	Interactive Graph Visualization with Zoom
Module	Data Visualization and Exploration
Priority	High
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION
Advanced interactive graph visualization system for relationship networks that allows users to explore complex networks.

3. PROCESS

1. Initial graph loading:

* User accesses the visualization module
* System loads the graph according to parameters

4. OUTPUTS

* Interactive visualization canvas/viewport

5. SOURCE
Input data

6. RESTRICTIONS
Scalability: 20–40 nodes

RF-019: GRAPH REFRESH BUTTON

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-VIS-002
Name	Graph Refresh Button
Module	Data Visualization and Exploration
Priority	High
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION
The system incorporates an interactive button that allows the user to refresh or update the visualized graph in real time, incorporating changes in the underlying data.

3. PROCESS

1. The user accesses the visualization module and selects the refresh button.

2. The system queries the updated data sources (e.g., new detections or mappings).

3. The graph is reloaded in the canvas/viewport, preserving previous configurations such as zoom or filters.

4. OUTPUTS

Updated graph in the interactive interface, with possible notifications of changes (e.g., "Update completed: X nodes modified").

5. SOURCE
Input data from RF-013 and RF-014

6. RESTRICTIONS
Scalability limited to 20–40 nodes

RF-020: FILTERS SECTION

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-VIS-003
Name	Filters Section
Module	Data Visualization and Exploration
Priority	High
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION
The system incorporates a dedicated interactive filters section that allows the user to refine the graph visualization.

3. PROCESS

1. The user accesses the visualization module and selects the filters section (e.g., via a side panel or dropdown).

2. The system presents filter options (e.g., by entity type, date range, transfer amounts, or presence in lists).

3. When filters are applied, the system updates the graph in the canvas/viewport, reloading only the relevant elements.

4. OUTPUTS
Filtered graph in the interactive interface

5. SOURCE
Input data

6. RESTRICTIONS
Filters will be enabled only if there is normalized data available.

RF-021: LIFECYCLE TRACKING OF COLLECTED CASE DATA BY DAY, PDF TABLE FORMAT

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-REPORT-001
Name	Lifecycle Tracking of Collected Case Data by Day, PDF Table Format
Module	Reports and Data Export
Priority	High
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION
The system generates a lifecycle trace of the data collected in a specific case, organized by days, in exportable table format to PDF.

3. PROCESS

1. The user selects a case or dataset (e.g., from the visualization module).

2. The system collects and sorts the data by date/day, tracing lifecycle stages (e.g., ingestion, filtering, detection, mapping).

3. A structured table is generated and exported to PDF, applying any available filters.

4. OUTPUTS
PDF file containing chronological tables of the data lifecycle

5. SOURCE
Input data

6. RESTRICTIONS
Limited to cases with manageable, relevant data volumes.

RF-022: TRACKING BY NAMES

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-REPORT-002
Name	Tracking by Names
Module	Reports and Data Export
Priority	High
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION
The system generates a detailed trace of entities, connections or patterns associated with specific names.

3. PROCESS

1. The user selects a name or set of names (e.g., from the visualization module or filters).

2. The system collects and organizes the data linked to those names, tracing connections, transfers or appearances in lists.

3. A structured table organized by name is generated and exported to PDF, applying any available filters.

4. OUTPUTS
PDF file with tables organized by names

5. SOURCE
Input data and RF-021

6. RESTRICTIONS
Limited to traces with manageable data volumes

RF-023: MONITORING OF UNUSUAL ACCELERATIONS IN BUREAUCRATIC DEADLINES

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-DETECT-002
Name	Monitoring of Unusual Accelerations in Bureaucratic Deadlines
Module	Financial Analysis and Anomaly Detection
Priority	Critical
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION
The system monitors and detects unusual accelerations in bureaucratic process deadlines by comparing captured dates.

3. PROCESS

1. The system collects bureaucratic deadline data from relevant sources (e.g., procedure registries).

2. An algorithm is applied to compare current deadlines against historical or expected durations.

3. Results are cross-referenced with previous detections (e.g., close circles or closed groups) and alerts are generated if correlated with suspicious entities.

4. OUTPUTS

Start and end dates of detected accelerated processes

5. SOURCE
RF-013 for the data to be used

6. RESTRICTIONS
Dependent on availability of data: start and end dates detected

RF-024: HISTORICAL RECORD OF DATA COLLECTIONS BY BATCH/PACKAGE

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-AUDIT-001
Name	Historical Record of Data Collections by Package
Module	Data Audit and Traceability
Priority	Critical
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION
The system must maintain a complete and auditable historical record of every data collection performed in batches/packages.

3. PROCESS

1. Every time a data collection is executed (manual or scheduled), the system creates a new "package".

2. All package metadata is automatically recorded.

4. OUTPUTS
Historical table containing the main fields:

* Package ID
* Start / end date and time

5. SOURCE
Module 1

6. RESTRICTIONS
Use of a simple relational database.

RF-025: AUTOMATIC COMPARISON BETWEEN SUBMITTED BIDS AND THE WINNING BID, BY FILE NAME

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-ANALYZE-001
Name	Automatic Comparison between Submitted Bids and the Winning Bid, by File Name
Module	Financial Analysis and Anomaly Detection
Priority	Critical
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION
The system performs an automatic comparison between the bids submitted in a process (e.g., tender or procurement) and the winning bid, by price, using the collected data.

3. PROCESS
The user queries the RTL records and PDFs to perform a manual inspection.

4. OUTPUTS
PDF generated as in RF-021

5. SOURCE
Processed data

6. RESTRICTIONS
Manual inspection required

RF-026: TRACING OF IDENTIFIED BENEFICIARIES IN THE NETWORK GRAPH

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-REPORT-003
Name	Tracing of Identified Beneficiaries in the Network Graph
Module	Reports and Data Export
Priority	High
Status	In definition
Version	1.0
Date	05/02/2026

2. DESCRIPTION

The system generates a trace of the beneficiaries identified in the network graph (e.g., people or entities that receive benefits in contexts of transfers, closed groups or detected anomalies), organized by their connections in the graph.

3. PROCESS

The system takes the filtered data and prepares it for visualization.
The user uses Module 3 to view the historical trace.

4. OUTPUTS

Interactive graph

5. SOURCE

Input data

6. RESTRICTIONS

Visualization limited by blocks / chunks

RF-027: TRACING OF BUDGETARY TRANSFERS BETWEEN GOVERNMENT AGENCIES

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-TRACK-004
Name	Tracing of Budgetary Transfers Between Agencies
Module	Module 3: Process Tracking (and Module 1: Data Listener)
Priority	High
Status	To be defined
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must allow tracing and monitoring of financial flows (budgetary transfers) that occur between different government agencies or public entities.

3. PROCESS

1. Ingestion: Module 1 (Data Listener) captures official records of budgetary movements (transfers, reallocations, fund releases) from official sources or financial APIs.

2. Processing: The system links each transaction to the sending and receiving agency. Module 2 (Network Analysis) is used to establish "finances" or "transfers_to" relationships between institution nodes.

3. Tracing: Module 3 organizes this data chronologically

4. OUTPUTS

* Timeline visualization

5. SOURCE

* Input data

6. RESTRICTIONS

* Nomenclature Matching: The system must standardize agency names

RF-028: APPROVAL CHAIN TRACING

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-TRACK-005
Name	Tracing of Approvals and Signature Chain
Module	Module 3: Process Tracking
Priority	High
Status	In definition
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must generate a sequential and immutable trace of all approvals and validations associated with a file or administrative process.

3. PROCESS

1. Metadata Extraction

Chronological Ordering: Module 3 sorts approval events sequentially (Step 1 → Step 2 → Step N), based on timestamps

4. OUTPUTS

* Approval Chain Visualization

5. SOURCE

* Input data

6. RESTRICTIONS

Only if the APIs provide this information

RF-029: PREPARATION OF INFORMATION IN COMPREHENSIVE TIMELINES (MODULE 3)

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-VISUAL-006
Name	Preparation of Information in Comprehensive Timelines (Module 3)
Module	Module 3: Process Tracking
Priority	High
Status	In definition
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must consolidate and structure all processed information within Module 3.

3. PROCESS

1. Data Aggregation: The system collects scattered events from Module 3: transfer dates (inflows/outflows), approval records, official document timestamps, and anomaly alerts.

2. Temporal Normalization: All date and time formats from different sources are standardized to a common timeline.

3. Layer Classification: Information is organized into categories or "lanes" within the timeline (e.g., Financial Layer, Administrative Layer, Legal/Beneficiaries Layer) to avoid visual overload.

4. Synchronization: Related events are linked

4. OUTPUTS

* Interactive Timeline

5. SOURCE

* Input data

6. RESTRICTIONS
Performance: The timeline must efficiently render hundreds of events without crashing the browser.

RF-030: VISUAL TIMELINE PER PROCESS WITH HIGHLIGHTED ANOMALIES

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-VISUAL-007
Name	Visual Timeline per Process with Highlighted Anomalies
Module	Module 3: Process Tracking (Visualization)
Priority	High
Status	In definition
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must generate a specific visual timeline for each selected process or file, graphically highlighting any detected deviation, irregularity or suspicious behavior.

3. PROCESS

1. Process Selection: The user selects a specific file or case within Module 3.

2. Interactive Detail: When interacting with a highlighted anomaly, a card is displayed with the technical justification of why it is considered a risk.

4. OUTPUTS

* Interactive Timeline

5. SOURCE

* Processed input data, RF-029

6. RESTRICTIONS

RF-029 must be operational

3.4 Non-Functional Requirements Definition

RNF-001: Data Security - All collected and processed data must be encrypted at rest and in transit

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-SEC-001
Name	Encryption of Data at Rest and in Transit
Module	Security and System Architecture (Cross-cutting)
Priority	Critical
Status	Defined
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must guarantee the protection of sensitive information collected and processed.

3. PROCESS

1. Definition of Policies: Establishment of mandatory encryption standards

2. Encryption in Transit: Configuration of SSL/TLS certificates on all API endpoints and enabling secure protocols (HTTPS).

4. OUTPUTS

Encrypted data

5. SOURCE

Functional requirements

6. RESTRICTIONS

Do not use obsolete or considered insecure algorithms

RNF-002: End-to-End Encryption (E2EE)

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-SEC-002
Name	End-to-End Encryption (E2EE)
Module	Security and Communications (Cross-cutting)
Priority	Critical
Status	In definition
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must implement an end-to-end encryption (E2EE) scheme for all sensitive information transmitted between system users or stored on their local devices.

3. OUTPUTS

* Secure communications

4. SOURCE

Modules 1 to 5

5. RESTRICTIONS

Key Recovery: As it is E2EE, if the user loses their private key or password, data recovery is not possible.

RNF-003: Every captured, analyzed or exported data item must have a SHA-512 hash and certified timestamp

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-SEC-003
Name	Integrity and Digital Chain of Custody (SHA-512 Hash + Certified Timestamp)
Module	Security and Forensic Audit (Cross-cutting)
Priority	Critical
Status	Defined
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must ensure the immutability and forensic integrity of all processed information by automatically generating a SHA-512 cryptographic hash and a certified timestamp (time seal) for each block of information.

3. PROCESS

1. Capture and Hashing: At the exact moment a data item enters the system (API, manual file or listener), its digital fingerprint is calculated using the SHA-512 algorithm.

4. OUTPUTS

* Integrity Record

5. SOURCE

* Input data

6. RESTRICTIONS

Standard Algorithm: Use of SHA-512 is mandatory

RNF-004: Operational Anonymity of the Investigator - Automatic IP Rotation

1. GENERAL INFORMATION

Field	Detail

Requirement ID	REQ-SEC-004
Name	Operational Anonymity of the Investigator and Automatic IP Rotation
Module	Security and Network (Infrastructure / Proxy)
Priority	High
Status	In definition
Version	1.0
Date	24/05/2024

2. DESCRIPTION
The system must guarantee the protection of the physical and digital identity of the investigators using the platform.

3. PROCESS
Automatic IP Rotation: The system must assign a different IP address for each session, query or predefined time interval (e.g. every 5 minutes), randomly selected from a global pool of IPs.

4. OUTPUTS

* Anonymized traffic

5. SOURCE
Connection device

6. RESTRICTIONS
Quality of the IP Pool: The IPs used must be of high reputation to avoid request blocking or other issues.

RNF-005: Automatic Detection and Blocking of Tracking Attempts

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-SEC-005

Name Automatic Detection and Blocking of Tracking Attempts (Anti-Tracking)
Module Security and Identity Protection (Client Layer and Proxy)
Priority High
Status In definition
Version 1.0
Date 24/05/2024

2. DESCRIPTION
The system must implement proactive digital defense mechanisms to detect and neutralize tracking techniques used by third parties.

3. PROCESS

1. Blocking of Unsolicited Telemetry: Filter and discard any incoming packet that does not match the expected API.

4. OUTPUTS
Secure channel

5. SOURCE

* Input data

6. RESTRICTIONS
Resource consumption – basic optimization required

RNF-006: Offline Availability

1. GENERAL INFORMATION
Field Detail
Requirement ID REQ-INFRA-007
Name Offline Operation Mode (No Internet Availability)
Module Infrastructure and Client (Mobile and Desktop App)

Priority	Medium
Status	In definition
Version	1.0
Date	24/05/2024

2. DESCRIPTION
The system must allow continuous operation of the application on mobile and desktop devices even in the complete absence of an internet connection.

3. PROCESS

1. Intelligent Local Cache: Securely store data on the device (encrypted local database).

4. OUTPUTS

* Functional interface without network

5. SOURCE

* Input data

6. RESTRICTIONS

* Storage Capacity: The size of the local database must be managed to avoid filling the user's storage.

RNF-007: Scalability for Large Cases

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-SCAL-008
Name	Scalability for Handling Large Cases (Big Data)
Module	Infrastructure and Backend Architecture
Priority	High
Status	In definition

Version	1.0
Date	24/05/2024

2. DESCRIPTION
The system must be capable of managing, processing and visualizing massive volumes of data associated with high-level investigations (large cases) without significant performance degradation.

3. PROCESS
Clean separation of perfectly scalable modules.

4. OUTPUTS

* Resource Metrics: Monitoring of CPU, RAM and Latency demonstrating that the system scales linearly with increasing traffic or data volume.

5. SOURCE

* Input data

6. RESTRICTIONS
Client-side hardware limitations

RNF-008: Mobile Interface

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-UI-010
Name	Adaptive User Interface for Mobile Devices (Mobile UI)
Module	Frontend / User Experience (Mobile)
Priority	High
Status	In definition
Version	1.0
Date	24/05/2024

2. DESCRIPTION
The system must feature a native or hybrid mobile application interface optimized specifically for small touch screens (smartphones and tablets).

3. PROCESS

1. Responsive and Adaptive Design: Implementation of a "Mobile-First" design where UI elements fluidly reorganize

Abstraction of Complexity (Visual Simplification):

* Graphs (Module 2): Instead of displaying the full graph (unreadable on mobile), the interface should offer a "Radar View" or list of key connections

4. OUTPUTS

* Functional Mobile Application

5. SOURCE
App

6. RESTRICTIONS
Unlike PC, mobile cannot display dozens of data columns simultaneously. Implementing filter systems is mandatory.

RNF-009: Dashboard Designed for Analysts

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-UI-011
Name	Analyst Control Dashboard (Analyst Workspace)
Module	Frontend / User Experience (Command Center)
Priority	High
Status	In definition
Version	1.0
Date	24/05/2024

2. DESCRIPTION
The system must provide a centralized and customizable "Dashboard" interface specifically designed for the workflow of an intelligence analyst.

3. PROCESS
Allow the analyst to add, remove and rearrange "widgets" or visual components in their main workspace according to their specific needs.

4. OUTPUTS

* Main Command Interface

5. SOURCE
App

6. RESTRICTIONS

* Information Density: Unlike mobile, analysts on PC typically use large screens (or dual monitors). The dashboard should take advantage of screen width to display more information in parallel (e.g., alerts list on the left and graph summary on the right) instead of a single narrow column layout.

RNF-010: Protection Against Evidence Tampering Using SHA Hash

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-SEC-012
Name	Protection Against Evidence Tampering Using SHA Hash
Module	Security and Chain of Custody
Priority	Critical
Status	Defined
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must guarantee the immutable integrity of all collected digital evidence by verifying the SHA hash (SHA-256 or higher) of relevant data through a verification button.

3. PROCESS

1. The user clicks on "Verify".

2. The system displays a message.

4. OUTPUTS
Message indicating whether there is a problem (tampering detected or integrity confirmed)

5. SOURCE
Stored data

6. RESTRICTIONS
There must be stored data with its corresponding hash

RNF-011: The system must self-destruct if it detects that it is being analyzed in a virtual or debugging environment

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-SEC-013
Name	Self-Destruction upon Detection of Virtual or Debugging Environment (Anti-Forensics)
Module	Endpoint Security / Application Protection (PC and Mobile)
Priority	Critical
Status	Sensitive / In definition
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must implement an active defense mechanism (Kill Switch) that continuously monitors the application's execution environment. If indicators are detected that the software is running inside a virtual machine (VM), emulator, sandbox environment, or if a debugger is attached to the process, the system must activate a self-destruct protocol.

3. PROCESS

1. Environment Detection (Environment Check):

* On PC: Check for typical virtualization artifacts (specific VM video drivers, virtual network card MAC addresses, presence of hypervisor services, constant CPU temperature, forcibly synchronized system clock).

* On Mobile: Detect if the device is rooted/jailbroken.

4. OUTPUTS

* Session Termination

5. SOURCE

List of running processes

6. RESTRICTIONS

* Local Data Loss: Self-destruction implies the complete loss of all information. The user must be aware that running in unsafe environments results in the annihilation of local data.

RNF-012: Code Obfuscation

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-SEC-014
Name	Code Obfuscation and Source Code Protection
Module	Development Security (DevSecOps)
Priority	High
Status	Defined
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must implement rigorous code obfuscation techniques across all layers of the application.

3. PROCESS

Semantic Renaming: Replace names of variables, functions, classes and parameters in the compiled code with meaningless, random or short character strings.

4. OUTPUTS

Obfuscated binaries

Mapping files (Map Files)

5. SOURCE

Application source code

6. RESTRICTIONS

Once obfuscated, error reports (Crash Reports) sent to servers will be unreadable unless processed with the private mapping files.

RNF-013: Protection against memory dumping

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-SEC-015
Name	Protection against Memory Dumping (Anti-Memory Dumping)
Module	Application Security (Runtime Protection)
Priority	High
Status	Defined
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The objective is to prevent attackers from performing a "memory dump" to recover information.

3. PROCESS

Anti-Dump (Fooling the Dumper):

Modify the executable headers in memory (PE Header / Mach-O) to break the structure expected by dumping tools, causing the generated dump file to be corrupted or unable to be opened by standard analyzers.

4. OUTPUTS

Protected memory

5. SOURCE

Memory

6. RESTRICTIONS

Performance Overhead: Constantly encrypting and decrypting data in memory consumes additional CPU cycles. This protection should be applied selectively only to the most critical data (secrets).

RNF-014: Daily renewal of encryption keys

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-SEC-016
Name	Daily Rotation of Encryption Keys (Key Rotation)
Module	Cryptographic Key Management (KMS)
Priority	High
Status	In definition
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must implement an automatic cryptographic key rotation policy.

3. PROCESS

Automatic Key Generation (Key Scheduling): A secure service (or the KMS - Key Management Service module) must automatically generate a new key pair or symmetric key daily without manual intervention. The old key set must be immediately archived.

Progressive Re-encryption (Data Re-encryption):
For new data: Use the keys of the "current day" (Day N) for encryption.
For existing data (encrypted with Day N-1 keys): The system must initiate a background process to decrypt records with the old key and re-encrypt them with the new key, or maintain an envelope encryption system where the daily "master key" protects the data keys.

4. OUTPUTS

* New Daily Key

5. SOURCE

System clock (UTC)

6. RESTRICTIONS

Only for important data.

RNF-015: Screen lock if screen capture tools are detected

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-SEC-017
Name	Screen Lock upon Detection of Capture Tools (Anti-Screen)
Module	Interface Security (Frontend / OS Integration)
Priority	High
Status	In definition
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must actively monitor the application environment to detect the use of screenshot or video recording tools.

3. PROCESS

1. Operating System Event Monitoring:

* On PC (Windows/macOS): Listen for operating system events related to screenshots (e.g. Print Screen key, Win+Shift+S). Check if a recording window (such as OBS Studio, Teams, Zoom) is active, overlaid or capturing the application area.

4. OUTPUTS

* Protected Screen

5. SOURCE

Application

6. RESTRICTIONS

Detection must be highly accurate to avoid false positives.

RNF-016: Dummy traffic to hide communication patterns

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-SEC-018
Name	Generation of Dummy Traffic to Obfuscate Patterns
Module	Network Security and Communications
Priority	High
Status	In definition
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must implement a mechanism for continuous and random generation of fictitious network traffic (dummy traffic).

3. PROCESS

1. Injection of Filler Packets (Cover Traffic): The application must generate encrypted data packets of variable sizes and send them periodically.

4. OUTPUTS

* Obfuscated network flow

5. SOURCE

API requests

6. RESTRICTIONS

* Bandwidth and data consumption

RNF-017: Programmable self-destruction

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-SEC-019
Name	Programmable Self-Destruction Mechanism (Dead Man's Switch)
Module	Security and Data Lifecycle Management
Priority	High
Status	In definition
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must incorporate a critical security function that allows programming the irreversible destruction of the application and all sensitive data stored on the device after a period of inactivity.

3. PROCESS

1. Timer Configuration:

* The user or administrator can define a countdown timer

In the "Dead Man's Switch" variant, the timer is automatically reset every time the user successfully authenticates.

4. OUTPUTS

* Clean device (data wiped)

5. SOURCE

Absence of the investigator

6. RESTRICTIONS

Irreversible data loss

RNF-018: "Reverse honeypot" mode – The system can generate fake data or misleading signals if it detects intrusion attempts

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-SEC-020
Name	Reverse Honeypot Mode (Active Disinformation)
Module	Offensive Security / Active Defense
Priority	High
Status	In definition
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must incorporate an active defense mechanism ("Reverse Honeypot") that, upon detecting an unauthorized intrusion attempt, forensic analysis, or environment manipulation, stops presenting real data and begins generating and serving fake data instead.

3. PROCESS

1. Compromise Detection: The system monitors the same conditions as the self-destruct requirement.

2. Generation of Synthetic Data

4. OUTPUTS

* "Zombie" interface (fake data presentation)

5. SOURCE

Uses criteria from RNF-017 and RNF-016

6. RESTRICTIONS

* Data Coherence: Fake data must be mathematically and logically consistent

RNF-019: Internal audit of the investigator's activity

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-AUD-021
Name	Internal Audit of Investigator Activity
Module	Governance, Compliance and Security
Priority	High
Status	In definition
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must implement a comprehensive logging and monitoring mechanism of all actions performed within each case.

3. PROCESS

The system captures login and logout events (and other relevant actions).

4. OUTPUTS

Array / log containing access records

5. SOURCE

User

RNF-020: API encryption

1. GENERAL INFORMATION

Field	Detail
Requirement ID	REQ-SEC-022
Name	API Communications Encryption (TLS/Cryptography)
Module	Communications Security and Backend
Priority	Critical
Status	Defined
Version	1.0
Date	24/05/2024

2. DESCRIPTION

The system must ensure that all data communication between the client (Mobile/PC Application) and the server, as well as between internal microservices, is carried out exclusively through encrypted channels.

3. PROCESS

Security Enforcement (HSTS): Configuration of HTTP Strict Transport Security (HSTS) headers on the server to force clients to always use encrypted connections (HTTPS).

4. OUTPUTS

* Encrypted traffic

5. SOURCE

Network traffic

6. RESTRICTIONS

Asymmetric encryption should only be used during the handshake (key exchange).